Cambridge Secondary 1 Checkpoint

Cambridge Assessment International Education
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CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

5346234123

SCIENCE **1113/01**

Paper 1 **October 2019**

45 minutes

Candidates answer on the Question Paper.

Additional Materials: Pen, Pencil, Ruler, Calculator

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
You should show all your working in the booklet.

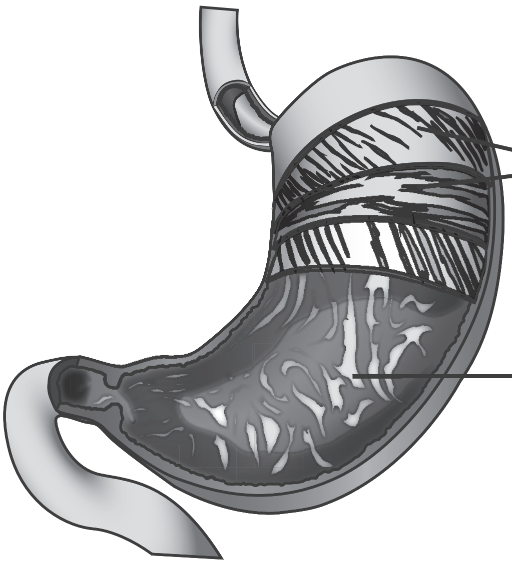
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 50.

This document consists of **14** printed pages and **2** blank pages.

IB19 10_1113_01/3RP

Cambridge Assessment International Education

1 The diagram shows a human stomach.



(a) Which term best describes the stomach?

Circle the correct answer.

cell **organ** **organism** **system** **tissue**

Use information from the diagram to explain your answer.

....

....

[2]

(b) Red blood cells and muscle cells are found in the wall of the stomach.

(i) Explain how the structure of a red blood cell is related to its function.

....

.... [2]

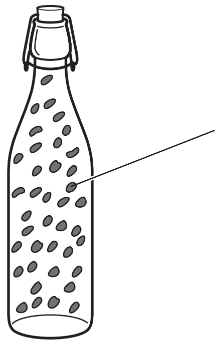
(ii) Explain how the structure of a muscle cell is related to its function.

....

.... [2]

2 Look at the diagram.

It shows a sealed bottle containing air particles.



(a) The air exerts a pressure on the sides of the bottle.

Explain how air exerts a pressure.

Use ideas about particles in your answer.

....

.... [1]

(b) The bottle is heated.

What happens to the pressure inside the bottle? ....

Explain your answer.

Use ideas about particles.

....

.... [2]

(c) A bottle of perfume is left in a room.

The top is taken off the bottle.

The smell of the perfume eventually fills the room.

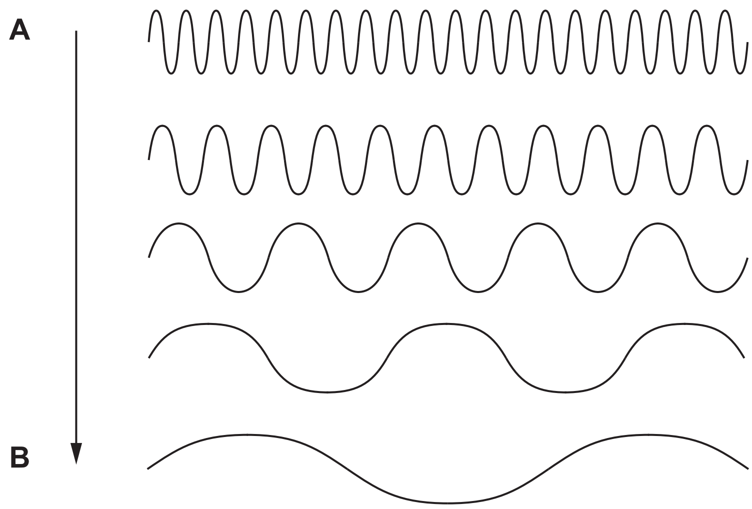
What is the name of this process?

Circle the correct answer.

condensation **conduction** **diffusion** **melting**

[1]

3 Chen looks at different sound traces with an oscilloscope.



Describe what is happening to the sound from **A** to **B**.

Choose words from

decreases **increases** **stays the same**

The pitch of the sound ....

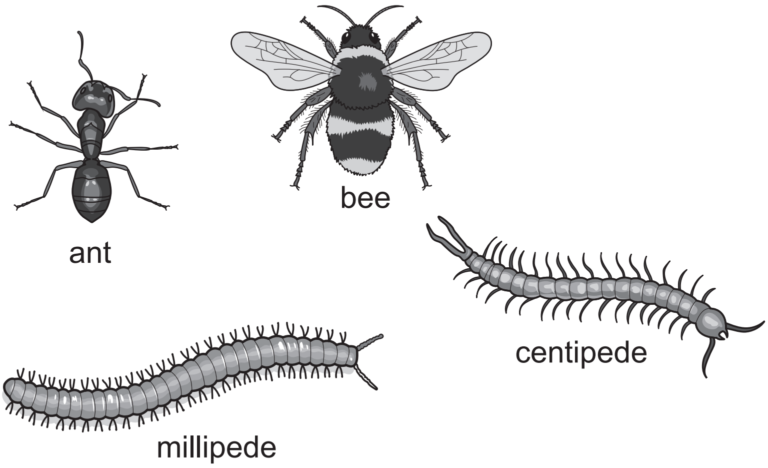
The frequency of the sound ....

The volume of the sound ....

The amplitude of the sound ....

[4]

4 This question is about completing a key to identify these four arthropods.

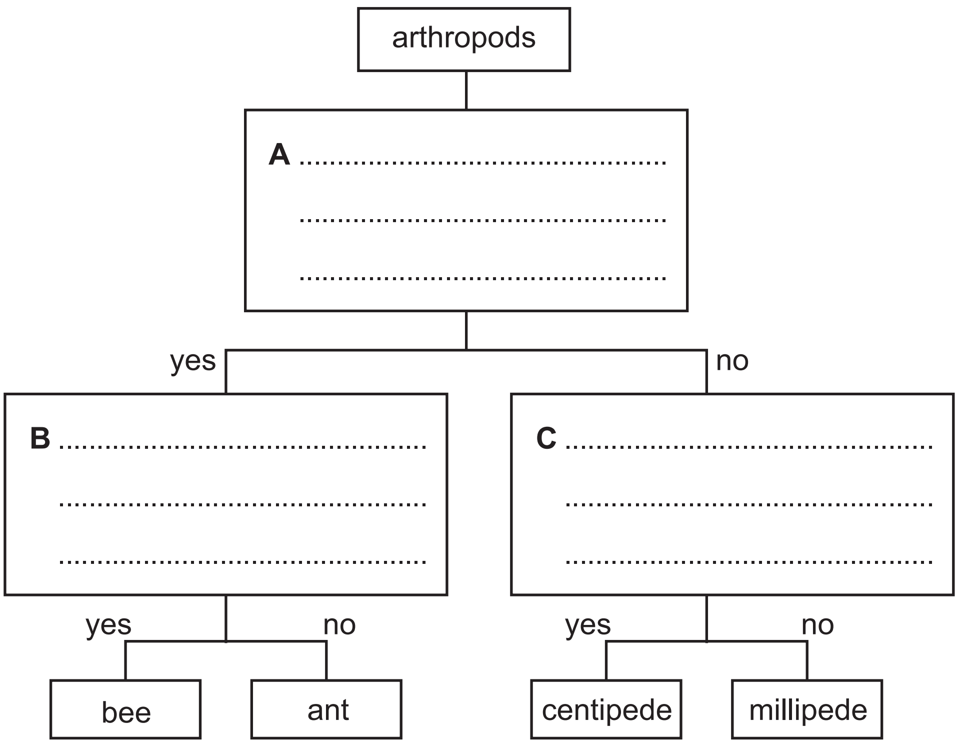


not to scale

(a) Look at the key.

The statements for boxes **A**, **B** and **C** are missing.

Use the diagrams to complete **A**, **B** and **C**.



[3]

(b) Spiders also belong to the arthropod group.

Give **one** feature which separates a spider from the other four arthropods.

.... [1]

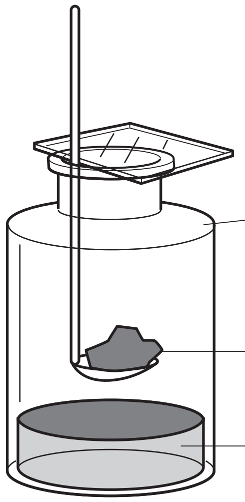
5 Blessy investigates the reactions of the element carbon.

Blessy uses a Bunsen burner to heat carbon.

She lets the carbon burn in air.

She then puts the burning carbon into a jar of oxygen.

The diagram shows the apparatus Blessy uses.



(a) Carbon is an element.

What is the chemical symbol for carbon? .... [1]

(b) The carbon reacts with the oxygen to make a gas.

(i) What is the name of this gas?

Circle the correct answer.

carbon carbonate **carbon dioxide** **carbon hydroxide** **carbon sulfate**

[1]

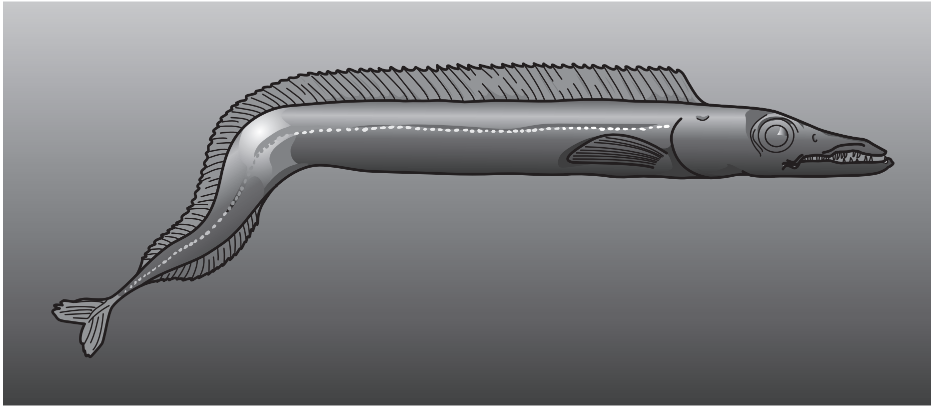
(ii) The gas turns Universal Indicator solution orange.

Suggest a pH value for the solution.

pH .... [1]

6 The diagram shows a black scabbard fish from the Atlantic Ocean.

This fish is adapted to live in very deep water where there is very little light.



(a) Describe how this fish is adapted to live where there is very little light.

.... [1]

(b) The black scabbard fish is a fast moving predator.

(i) Suggest how the **colour** of this fish helps to make it a successful predator.

.... [1]

(ii) Describe two **other** adaptations that suggest that this fish is a fast moving predator.

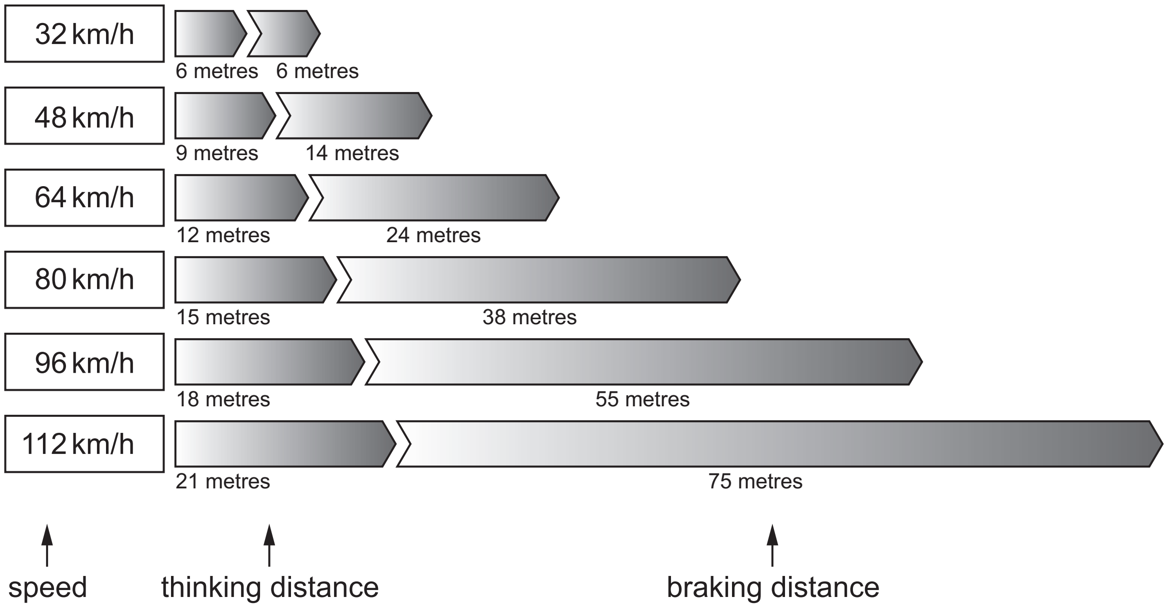
1 ....

2 ....

[2]

7 Angelique finds information about the stopping distance of cars.

The stopping distance is the distance a car moves after the driver wants to stop the car.



Angelique calculates the stopping distance for a car with a speed of 32 km/h.

Here is her calculation.

6 m + 6 m = 12 m

(a) Calculate the stopping distance for a car with a speed of 112 km/h.

stopping distance = ... m [1]

(b) Describe the pattern in the information for **thinking distance**.

Complete the sentence.

As the speed increases by 16 km/h the thinking distance ....

....

.... [2]

(c) Predict what the **thinking distance** will be at 128 km/h.

.... m [1]

(d) The speed of the car doubles from 32 km/h to 64 km/h.

Answer the questions choosing words from

decreases

doubles

halves

more than doubles

more than halves

What happens to the **thinking distance** from 32 km/h to 64 km/h?

....

What happens to the **braking distance** from 32 km/h to 64 km/h?

....

What happens to the **stopping distance** from 32 km/h to 64 km/h?

....

[2]

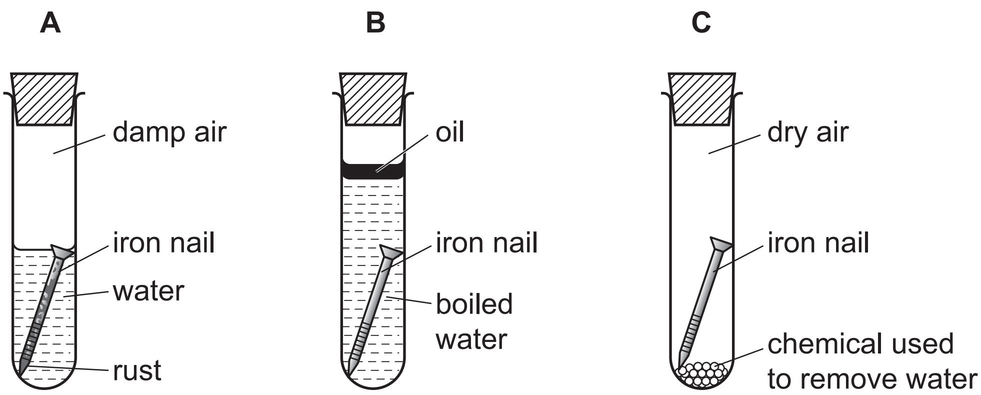
8 Safia investigates the rusting of iron.

She puts iron nails into three different test-tubes.

Each test-tube contains different conditions.

The test-tubes are then left for one week.

The diagram shows the test-tubes after one week.



(a) The iron nail in tube **A** rusts.

This is because the nail reacts with water and a gas found in air.

What is the name of the gas?

.... [1]

(b) Complete the sentences about the investigation.

The iron nail in tube **B** did **not** rust because ....

....

The iron nail in tube **C** did **not** rust because ....

....

[2]

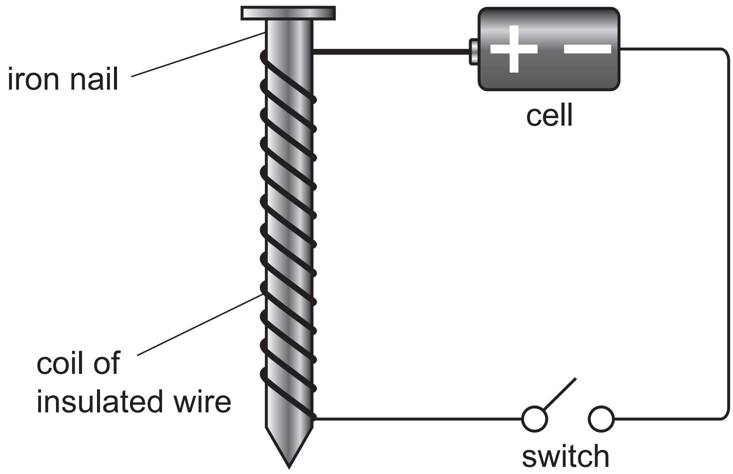
(c) Rusting is a reaction that is **not** useful.

Explain why rusting is **not** a useful reaction.

....

.... [1]

9 Pierre makes an electromagnet.



Pierre wants to make a **stronger** electromagnet.

What does he do?

Circle the **two** correct answers.

add another cell

add another switch

add more coils to the insulated wire

change the iron nail to a wooden pencil

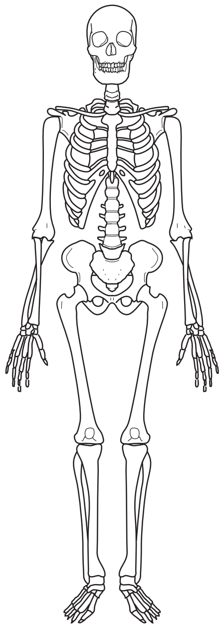
remove the iron nail

remove the switch

turn the cell around

[2]

10 Look at the diagram of a human skeleton.



(a) Name the tissue which makes up the skeleton.

.... [1]

(b) Give two **functions** of the skeleton.

1 ....

2 ....

[2]

11 Jamila investigates the reaction of different metals with hydrochloric acid.

The metals are magnesium, zinc and iron.

She measures how long it takes for the reaction with each metal to make 50 cm^3 of hydrogen gas.

(a) She thinks these variables are important in her investigation.

A the metal used

B the volume of hydrochloric acid used

C the concentration of the hydrochloric acid used

D the time to make 50 cm^3 of hydrogen gas

E the temperature of the room

Which letter shows the variable Jamila changes? ....

Which letters show the three variables Jamila keeps the same?

.... , ... and ....

Which letter shows the variable Jamila measures to find out which metal reacts the fastest?

.... [4]

(b) Jamila predicts that hydrochloric acid will react fastest with **zinc**.

Look at her results.

metal	**time to make 50 cm^3 of gas in seconds**
magnesium	50
iron	280
zinc	200

Is Jamila's prediction correct? ....

Explain your answer.

....

....

[2]

12 **(a)** Class 9 have a quiz about our solar system.

Complete the answers.

Solar System Quiz

1. Mercury, Earth and Mars are three of the inner planets of our solar system.

What is the name of the **other** inner planet?

....

2. The most distant planet from Earth **was** Pluto.

Pluto has **now** been classified as a dwarf planet instead of a planet.

What is the name of the **most** distant planet from Earth?

....

3. What is the name of the object that all the planets in our solar system orbit?

....

[2]

(b) Class 9 look at a photograph of the night sky.

A camera normally lets light into it for a second.

The camera that took this photograph let light into it for 30 minutes.

What objects in the night sky make the lines on the photograph?

....

Explain why they look like lines.

....

[2]

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